

Smart Traffic Congestion Prediction System

1. Introduction

This project aims to develop an intelligent system for predicting traffic congestion levels based on weather conditions and temporal factors. Traffic volume is classified into three categories: Low, Medium, and High. The goal is to help urban planners and transportation authorities make informed decisions to improve traffic flow and reduce commute times. The motivation stems from the growing challenge of urban traffic congestion, which leads to lost productivity, increased fuel consumption, and environmental pollution.

2. Dataset Description

The project uses the Metro Interstate Traffic Volume dataset containing approximately 48,000 hourly records. The dataset includes nine features: holiday indicator, temperature (Kelvin), hourly rainfall, hourly snowfall, cloud coverage percentage, main weather category, detailed weather description, date and time, and traffic volume. Traffic was categorized as: Low (under 2,000 vehicles), Medium (2,000-5,000 vehicles), and High (over 5,000 vehicles).

3. Models Used

Three machine learning algorithms were implemented for comparison: Logistic Regression served as a baseline linear classifier, Random Forest is an ensemble method using multiple decision trees, and K-Nearest Neighbors classifies traffic levels based on similar past records. All models were evaluated using four standard metrics: Accuracy, Precision, Recall, and F1-Score.

4. Neural Network

A deep learning model was designed with an input layer, two hidden layers (128 and 64 neurons respectively) using RELU activation function, dropout layers for regularization to prevent overfitting, and a softmax output layer for multi-class classification. The model was trained using the Adam optimizer with early stopping to prevent unnecessary training epochs. Training Accuracy reached approximately 91%.

5. Results & Comparison

<i>Model</i>	<i>Accuracy</i>	<i>Precision</i>	<i>F1-Score</i>
Logistic Regression	63.5%	59.2%	58.9%
KNN	75.3%	75.6%	75.5%
Neural Network	91.0%	91.0%	91.0%
Random Forest	93.0%	93.1%	93.1%

6. What I Learned

This project confirmed that traffic congestion can be predicted with reasonable accuracy using weather data and temporal features. Random Forest achieved the best performance with 93% accuracy. Key lessons include the importance of data cleaning and preprocessing, the impact of model selection on results, and that tree-based methods were more effective for this dataset. Future work could explore additional features such as road capacity, accident data, and special event information.